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Shallow seasonal stratification ameliorates coral bleaching during record-breaking marine heatwaves in a marginal subtropical system

Gonzalo Pérez-Rosales¹ · Yu-De Pei¹ · Joshua Bennett-Williams¹ · Timothy B. King¹ · Markus Rummel¹ · Tzu Hao Chung² · Alex S. J. Wyatt¹

Received: 30 January 2026 / Accepted: 9 February 2026
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Abstract Marine heatwaves are intensifying globally, with the Fourth Global Coral Bleaching Event (GCBE) beginning in 2023, yet coral bleaching outcomes remain highly variable across space and time. This study examines the role of fine-scale thermal variation in shaping bleaching impacts during two record-breaking marine heatwaves (2022 and 2024) affecting shallow coral assemblages in the marginal and urbanised environments of Hong Kong. Using high-resolution in situ temperature loggers, we quantified thermal exposure across coral-inhabited depths (1–8 m) at three sites and related this to bleaching prevalence across taxa. In 2024, surface temperatures exceeded the local bleaching threshold (29.7 °C) for 37 days, resulting in heat accumulation that reached 25.2 °C days (9.24 °C weeks) and remained above 8 °C days (°C weeks) for 46 days (20 days). However, seasonal stratification created vertical gradients of up to 9.6 °C between 2 and 8 m, reducing the magnitude and duration of heat accumulation severalfold (e.g., up to 35.4 °C days and 76 days > 8 °C days at 2 m, compared to 7.9 °C days at 8 m). Bleaching prevalence was 20–46% lower between 1 and 5 m, with Bayesian bleaching probabilities highest for *Goniopora* (82%) and lowest for *Porites* (64%). These patterns could not be explained by satellite-derived sea surface temperatures alone. Despite record heating and stress, mortality was limited (60 to 95% pigmentation recovery). Our findings provide rare fine-scale assessments of bleaching and thermal dynamics, including during the Fourth GCBE, and highlight how local stratification can create thermal refuges. Understanding how oceanographic contexts structure these refuges is essential for identifying and conserving coral communities with greater survival potential under accelerating ocean heating.

Supplementary Information The online version contains supplementary material available at <https://doi.org/10.1007/s00338-026-02835-w>.

✉ Alex S. J. Wyatt
wyatt@ust.hk

¹ Department of Ocean Science, The Hong Kong University of Science and Technology, Clear Water Bay, Kowloon, Hong Kong

² Department of Food Science and Nutrition, The Hong Kong Polytechnic University, Hung Hom, Hong Kong

Keywords Coral bleaching · Depth gradient · In situ temperature · Marginal community · Marine heatwave · Stratification

Introduction

The Fourth Global Coral Bleaching Event (GCBE) was declared following numerous worldwide reports of coral bleaching associated with prolonged marine heatwaves (NOAA, 2024; Reimer et al. 2024). According to NOAA Coral Reef Watch, up to 83.7% of the world's coral reefs were affected by bleaching (NOAA, 2025), which aligns

with 2024 setting another global record for ocean heat content (Cheng et al. 2025). The marginal and urbanised coral communities of Hong Kong (Cybulski et al. 2020; Duprey et al. 2016; Goodkin et al. 2011; Yeung et al. 2021) were no exception, with local news reporting significant bleaching observations (SCMP, 2024) expected to lead to more significant impacts than bleaching events from previous years (Chung et al. 2024; Wong et al. 2018; Xie et al. 2017). With significant seasonal temperature ranges, corals in Hong Kong frequently experience elevated seawater temperatures, reaching temperatures above the predicted local coral bleaching thresholds (i.e., based on NOAA definitions, Liu et al. 2018) occurring almost every summer in the past two

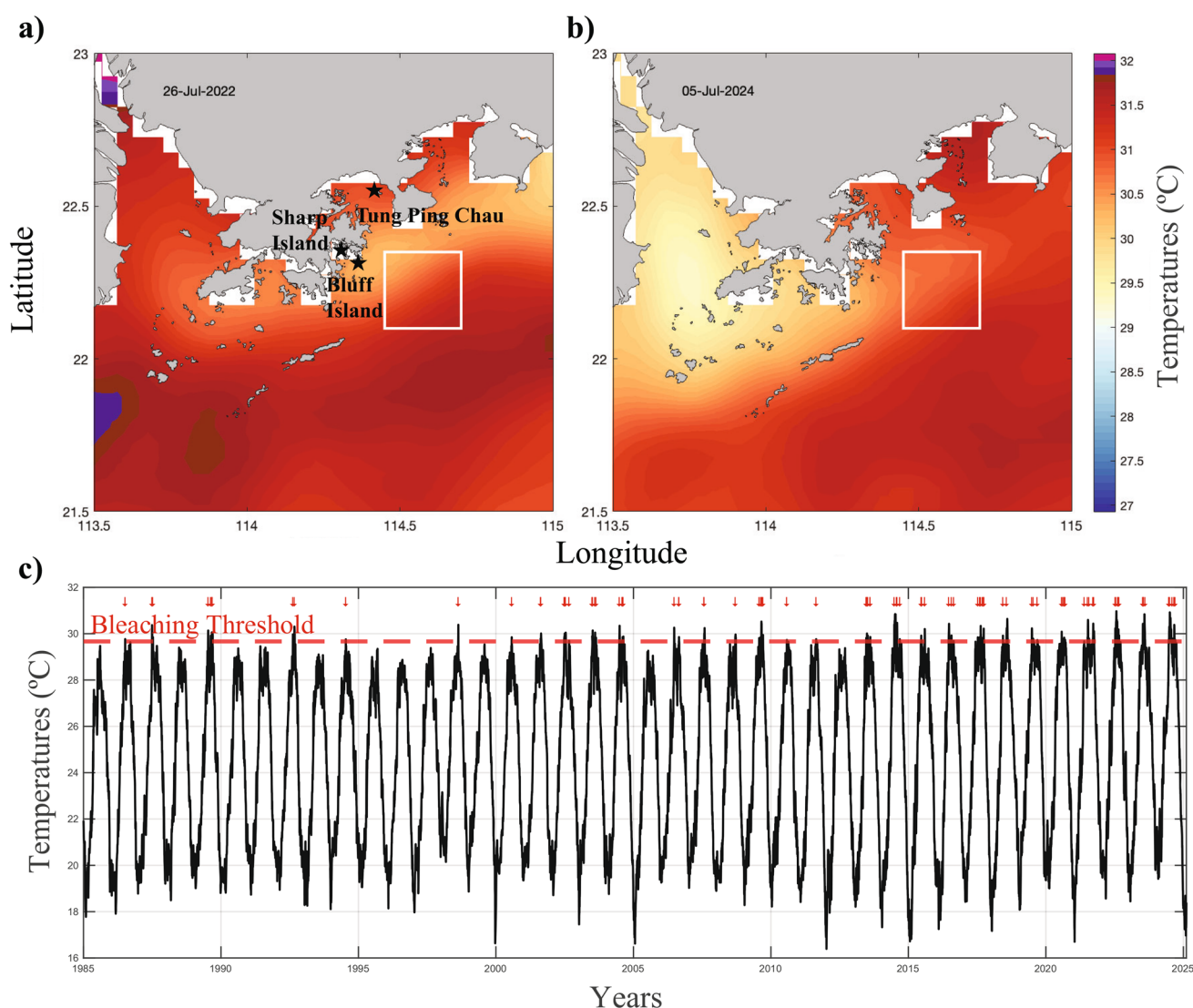


Fig. 1 Satellite sea-surface temperatures (SST) based on NOAA Coral Reef Watch (NOAA-CRW) data for the hottest days of (a) 26 July 2022 and (b) 5 July 2024. The three sites are indicated with a black star. The white boxes in (a) and (b) represent the grid pixels within which (c) a long-term SST time series was extracted from

1985 to 2025. The dashed line in (c) shows the Bleaching Threshold (BT, 29.67 °C), and the red arrows show the instances when SST exceeded the BT (i.e. when heat stress likely to trigger coral bleaching begins to accumulate)

decades (Chung et al. 2024; see results Fig. 1; Fig. S1). Yet, in nearly 40 years of data, Hong Kong ocean temperatures had never sustained such high temperatures for as long as in summer 2024 (Hong Kong Observatory 2024a). The NOAA Coral Reef Watch program issued a bleaching Alert Level 2 or 3 for the region on 6 August 2024 (NOAA-CRW, 2024). However, these satellite-derived sea surface temperatures may overlook critical subsurface thermal variability likely to shape bleaching outcomes, especially in marginal and highly dynamic systems like Hong Kong.

Despite marginal conditions and ongoing heating episodes, Hong Kong hosts ~90 scleractinian species and patchy coral communities persist (Duprey et al. 2016). Hong Kong coral communities are considered “marginal and urbanised”, experiencing strong seasonality and high levels of natural and anthropogenic stressors (Browne & Bauman 2023; Duprey et al. 2016, 2020; Goodkin et al. 2011; McIlroy et al. 2024). During the wet season (May–September), when over 80% of Hong Kong’s rainfall occurs, the southwestern monsoon pushes warmer waters shoreward, and seawater temperatures are around ~29 °C, with occasional highs (heatwaves) reaching > 30 °C. Conversely, during the dry season (October–April), the northeast monsoon prevails, cooling seawater temperatures down to ~16–18 °C, with occasional lows at ~13–14 °C (Chung et al. 2024; Goodkin et al. 2011; Morton 1994). Corals in Hong Kong thus persist under a thermal regime that is frequently outside typical thermal optimums (26–28 °C; Coles & Jokiel 1977; Hoegh-Guldberg 1999), although a recent global analysis has shown that shallow tropical and subtropical shallow waters exhibit more short-term thermal variability that may exacerbate physiological stress than previously assumed (Li Shing Hiung et al. 2024).

Besides strong thermal seasonality, Hong Kong’s oceanography is also very dynamic and spatially variable. The region is primarily influenced by nutrient-rich and turbid freshwater inflows from the Pearl River from the west and oceanic waters from the east (see Methods, Study Site). Along with freshwater inputs during frequent tropical storms (typhoons) during the wet season (Zhou et al. 2012), these different water bodies can lead to pronounced stratification with multiple marked thermoclines, even at shallow depths very close to the surface. Such stratification (i.e., water-column density layering with depth; commonly quantified using buoyancy frequency; Gregg 1987) is expected to be more pronounced in sheltered coastal areas where less mixing is expected than in areas of ocean waves and currents (Morton & Wu 1975). However, limited data is available to explore vertical patterns in water column structure (i.e., from regular conductivity, temperature, and depth (CTD) profiling across time) to understand and predict temporal and spatial changes. Hydrodynamic models can help (Gan

et al. 2016, Gan, 2021), but these require local validation. Although strong variations in thermal regimes around Hong Kong must influence coral distributions—particularly by driving bleaching patterns during heatwaves—temperature-bleaching relationships are not fully understood, especially when considering fine-scale, site- and depth-specific variability (Suggett & Smith 2020).

Here, we focused on the latest heatwave, which triggered coral bleaching around Hong Kong during the Fourth GCBE in 2024, to study the vertical (depth) and spatial (site-specific) patterns in bleaching prevalence and impacts according to high-resolution, in situ temperature data. By coupling exhaustive bleaching surveys during and after the 2024 heatwave with additional data from the 2022 event (Chung et al. 2024) and a holistic set of temperature observations (including satellite sea-surface temperatures (SST), high-resolution in situ fixed and camera-mounted logger data, and vertical CTD profiles), we aimed to (1) test whether decreasing temperatures with depth decrease coral bleaching; (2) quantify the extent to which decreasing temperatures and heat accumulation with depth may protect corals from bleaching; and (3) assess how this protection varies across key coral assemblages (i.e., specific sites) and between coral taxa (i.e., test genus-driven bleaching susceptibility). By integrating local oceanography and seasonal stratification, we aimed to assess how depth-driven cooling influences coral bleaching and mortality to inform conservation planning in this marginal setting.

Materials and methods

Study site

Hong Kong is located approximately 130 km south of the Tropic of Cancer in the subtropical region of the South China Sea, east of the Pearl River Estuary. The estuary is located in one of the world’s most urbanised regions, surrounded by large cities like Hong Kong, Shenzhen, Macau and Guangzhou (Duprey et al. 2020). Hong Kong comprises around 260 islands with a coastline of 1,100 km. Despite significant environmental challenges and anthropogenic stressors (Morton 1994; Duprey et al. 2016), Hong Kong’s coral communities are home to around 90 scleractinian species, over 50 octocorals, and over 370 reef-associated fish species (Duprey et al. 2016; AFCDC 2005; Chung et al. 2023; Pei et al. 2025). These coastal habitats and coral communities are subjected to high seasonal (i.e., wet season from around May to September and dry from October to April) and spatial variability, mainly influenced by the freshwater plume that leads to increasing estuarine conditions that decrease longitudinally from the west (Pearl River) to the east (oceanic waters) (Morton 1994; Yeung et al. 2021;

Gan et al. 2016). The coral communities, as well as other coral-associated fauna, follow the same environmental gradient (Cybulski et al. 2020; Duprey et al. 2016; Fabricius & McCorry 2006; Pei et al. 2025).

We focused on three sites known to have some of the highest coral cover in Hong Kong (Yeung et al. 2021) and where we have maintained in situ long-term, high-resolution water temperature observations since 2020 (Fig. 1). The three surveyed sites where we characterised bleaching events were Bluff Island (22.325144° N, 114.353753° E), near the oceanic opening east of Port Shelter connecting to Mirs Bay; Sharp Island (22.361253° N, 114.291753° E) in the semi-enclosed waters of inner Port Shelter; and Tung Ping Chau (TPC) (22.546694° N, 114.436472° E) more isolated from the coast but still within Mirs Bay.

Temperature context

Satellite sea surface temperatures (SST)

SST was obtained from the NOAA Coral Reef Watch (NOAA-CRW), which has a 5-km spatial resolution (0.05 degrees) and daily temporal resolution (Roberts-Jones et al. 2012; Skirving et al. 2020) (Fig. 1). We focused on SST values within a box off the coast of Hong Kong to avoid landmass effects (see Fig. 1; top left: 23° N and 113.5° E and bottom-right 21.5° N and 115° E). We extracted daily SST values from 1985 and calculated heat stress as Degree Heating Weeks (DHWs; °C weeks) and Degree Heating Days (DHDs; °C-days) above a Bleaching Threshold (BT) over 12 weeks and 12 days, respectively (Liu et al. 2018; Wyatt et al. 2023). The BT was based on the monthly (July) maximum mean (from 1985 to 1993) temperature plus one °C, calculated from the extracted SST time series as 29.67 °C, following NOAA standards (Liu et al. 2018). To compare the duration of heating events, we counted the number of days when the degree heating was above 8 °C weeks/days;

Table 1 Deployment summary and marine heatwave metrics

Site depth (m)	Instrument (Sampling Int.)	Dates	Max temp (°C)		Max DHD (°C days)		Duration DHD > 8 °C days (days)	
			2022	2024	2022	2024	2022	2024
SST 0	NOAA CRW (1 day)	01 Jan 1985–15 Apr 2025	31.97 26 Jul	30.92 5 Jul	23.1 3 Aug	25.2 16 Jul	31	46
<i>Bluff island</i>								
2	miniDOT (5 min)	10 May 2024–05 Nov 2024	NA	32.19 7 July	NA	22.0 26 Jul	NA	51
5	SBE56 (2 min)	04 Nov 2020–06 Nov 2024	31.93 15 Jul	31.65 23 Jul	4.08 2 Sep	15.44 25 Jul	0	20
5	miniDOT (5 min)	10 May 2024–05 Nov 2024	NA	31.6 23 Jul	NA	14.27 25 Jul	NA	17
<i>Sharp island</i>								
2	miniDOT (5 min)	28 Apr 2024–05 Nov 2024	NA	33.21 23 July	NA	35.41 17 Jul	NA	76
3	SBE56 (2 min)	18 Nov 2020–05 Nov 2024	33.15 15 Jul	32.9 10 Jul	18.54 26 Aug	25.09 25 Jul	39	64
4	SBE56 (2 min)	17 May 2021–03 Apr 2025	32.78 14 July	NA*	12.76 27 Aug	NA*	22	NA*
5	miniDOT (5 min)	28 Apr 2024–05 Nov 2024	NA	32.28 07 Jul	NA	19.15 25 Jul	NA	25
8	miniDOT (5 min)	28 Apr 2024–05 Nov 2024	NA	30.98 16 Jul	NA	7.90 24 Jul	NA	0
<i>Tung ping chau</i>								
3	SBE56 (2 min)	04 Nov 2020–06 Nov 2024	32.18 15 Jul	32.73 08 Jul	24.02 23 Jul	31.71 15 Jul	34	58

* 4 m logger at Sharp Island failed from 27 April to 12 November 2024

For the duration, the reference value of DHD > 8 °C days was used as a threshold to compare heat accumulation (see Methods)

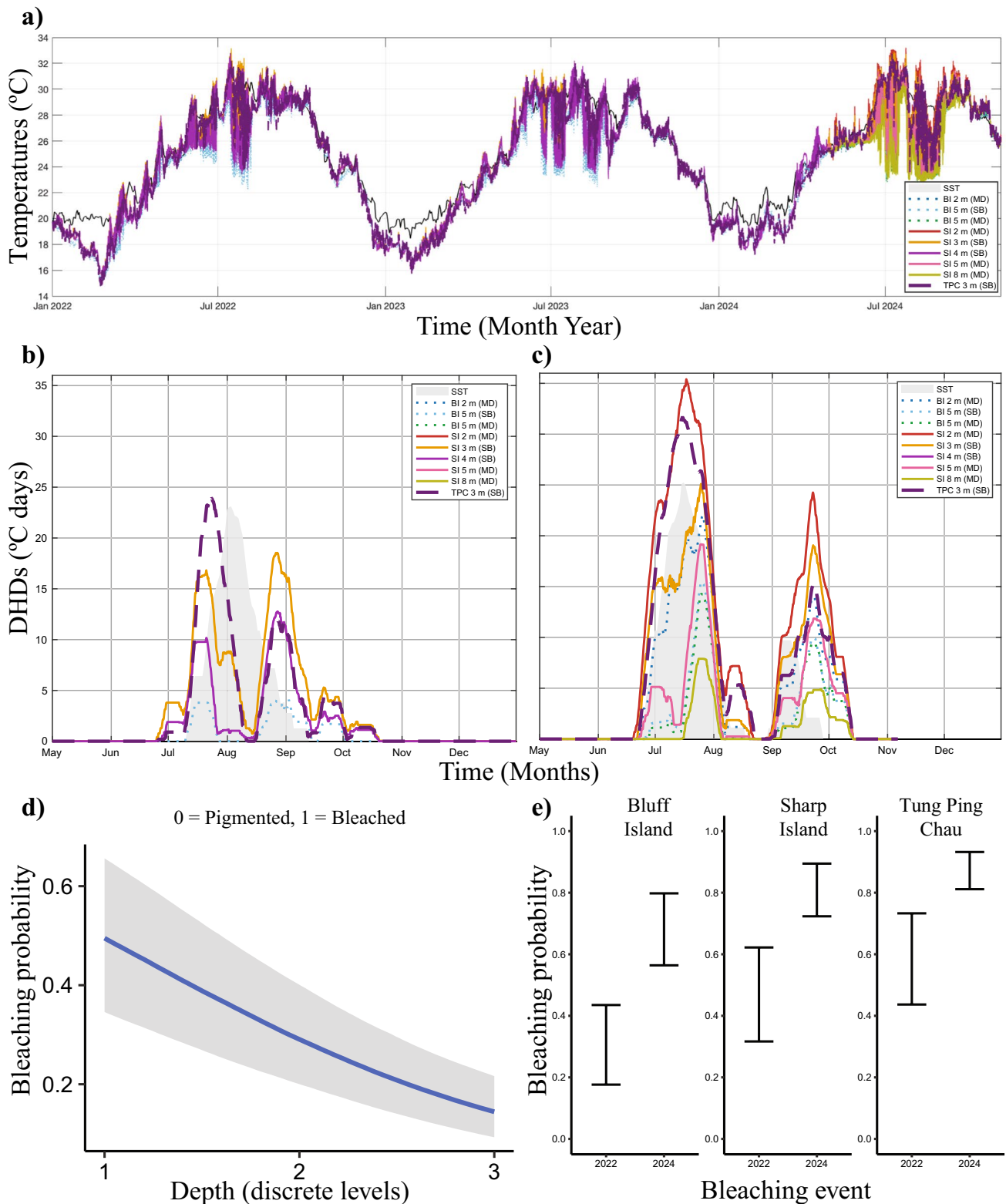


Fig. 2 Interactions between site and depth temperatures with bleaching probability. **a** Long-term in situ temperatures from a combination of logger deployments at different sites and depths (SBE56 (SB) and PME minidots (MD); see Table 1). **b** and **c** show the corresponding heat accumulation displayed as degree heating days (°C days) for

the **b** 2022 and **c** 2024 heatwaves, relative to a uniform Bleaching Threshold (BT) of 29.67 °C. Bayesian predicted bleaching probability (0=Pigmented, 1=Bleached) with conditional effects, according to **d** discrete depth levels (1=1 m (only 2024 data); 2=2–4 m, and 3=4–6 m), and **e** sites and heatwaves

a reference value often taken to imply expected widespread mortality (Liu et al. 2018).

Fixed in situ temperature loggers

We gathered high-resolution in situ water temperatures across depths at each site since November 2020, capturing the 2022 and 2024 bleaching heatwaves (Table 1). Instrumentation consisted of a combination of the SeaBird Electronics SBE56 thermistors (sampling at 2-min intervals at 5 m at Bluff Island, 3 and 4 m at Sharp Island, and 3 m at Tung Ping Chau), and Precision Measurement Engineering (PME) miniDOT loggers (sampling at 5-min intervals at 2 and 5 m at Bluff, and 2, 5 and 8 m at Sharp; Fig. 2). Instrument depths were determined in situ using a pressure sensor at the time of installation. Loggers were secured to fixed rebar posts anchored to the reef substrate, ensuring consistent vertical separation across deployments. Depths reflect nominal depths relative to instantaneous local sea level (i.e., not converted to Lowest Astronomical Tide) because local water variation at our sites cannot be reconstructed from regional tide stations. Minor differences in absolute depth (typically $< \sim 1$ m in this environment) do not affect analyses, which are based on relative differences rather than absolute chart-datum depth. For all temperature time series, we quantified heat accumulation as DHDs, as described in Wyatt et al. (2023). As previously mentioned, DHDs represented the accumulation of temperatures above the local BT derived from the regional SST as previously described and according to the standard methodology for coral heat stress assessment (Liu et al. 2018).

Vertical temperature profiles

In addition to fixed in situ loggers, we characterised water column hydrographic structure (vertical profiles) at least twice a year to capture seasonality at each site based on profiles taken with a Conductivity (salinity), Temperature and Depth probe (CTD; RINKO-profiler ASTD-102, JFE Advantech Co., Ltd., Tokyo, Japan). The CTD casts were performed along fixed onshore-offshore transects consisting of 10 stations 50 m apart, beginning close to shore (~ 1 m depth), and an additional station 500 m offshore to characterise the deeper water column structure. The depth of maximum stratification for each cast was determined from the depth of the maximum Brunt–Väisälä buoyancy Frequency squared (N^2) calculated as:

$$N^2 = -\left(\frac{g}{\rho}\right) \frac{\partial \rho}{\partial z}$$

where g is gravitational acceleration, ρ is density, and z is depth, using the “Gibbs-SeaWater Oceanographic” Toolbox

Table 2 Average depths and temperatures from the coral bleaching surveys undertaken at each of the three sites “During” the peak of bleaching in August 2024. Exact average values and standard deviations were obtained from the loggers DEFI2-D and DEFI2-T, respectively

Site	Date	Depths (m)	Temperatures (°C)
Bluff island	05 Aug	1.38 ± 0.25	29.86 ± 0.11
Bluff island	05 Aug	2.30 ± 0.13	28.27 ± 0.58
Bluff island	06 Aug	5.17 ± 0.2	24.46 ± 0.39
Sharp island	05 Aug	1.40 ± 0.26	31.02 ± 0.12
Sharp island	05 Aug	2.71 ± 0.29	29.34 ± 1.46
Sharp island	05 Aug	5.06 ± 0.24	25.16 ± 0.18
Tung ping chau	09 Aug	1.34 ± 0.34	31.56 ± 0.12
Tung ping chau	09 Aug	3.22 ± 0.24	29.49 ± 0.82
Tung ping chau	09 Aug	4.90 ± 0.22	25.06 ± 0.3

(McDougall and Barker, 2011). During 2022, CTD transects were realised in May and October at Sharp Island only. In 2024, CTD transects were taken at each of the three sites during the peak of the bleaching, coincident with the coral bleaching surveys in early August and after bleaching in November.

Bleaching surveys and coral-level temperatures and depths

At each site, we analysed nine 30 m transects deployed over available coral habitats as close as possible to the fixed loggers described above. At each depth, three transects were arranged continuously end-on-end at roughly 1, 2–3, and 4–5 m depths. The three transects per site and depth were carried out just after the peak of the bleaching (“During”, 5–6 and 9 of August 2024 at Sharp–Bluff islands and TPC, respectively) and three months after (“After”, 5 and 6 November 2024 at Sharp–Bluff islands and TPC, respectively) (Table 2). It should be noted that these cannot be considered as fixed transects, although we made every effort to ensure that the transects “After” the bleaching closely matched the coordinates and head directions of those “During” the bleaching. The transects consisted of videos of the community structure, filmed at 4 K resolution with a GoPro Hero 11 Black by a diver slowly swimming (~ 8 m/min) over the benthos, maintaining a fixed distance of 1 m above the reef. The fixed distance to the benthos was ensured by attaching the GoPro to a PVC frame that provided a 50 cm scale bar with coloured tape for colour correction. Two loggers were attached to the base of the camera frame to record high-resolution (1-s interval) data on depth and temperature (DEFI2-D and DEFI2-T, respectively, JFE Advantech Co., Ltd., Tokyo, Japan). Such loggers allowed the high-precision determination of transect-average depths and temperatures

at the actual level of the surveyed colonies (Table 2). These high-precision data were also used to test depth and temperature correlations with Pearson *cor.test* (package “stats”).

Posterior analyses consisted of identifying all coral colonies within the 0.5 m wide pole to the lowest taxonomic (i.e., based on morpho-taxonomy) level possible (morpho-type species later pooled to genus level; see text in Supplementary Information on the section “Quick Coral Id Guide”) and categorising them into the following statuses: “Pigmented” if presented with full natural colouration; “Partially Bleached” if bleaching was present but covered less than 50% of the colony or the colouration was considered to be pale; “Bleached” if over 50% of the colony was fully bleached or the colouration was white; and “Dead” if the colony presented signs of recent mortality and the dead part covered more than 50% of the colony. Pigmentation classifications were according to the referent colours of the most pigmented colonies, white balance calibrations of a white reference marker on the camera frame and the CoralWatch colour chart (Siebeck et al. 2006). All identifications and categorisations were done by the same observer (GP-R). To test bleaching patterns with depth and temperature across taxa during the peak of the bleaching (see [Bayesian modelling](#)), we transformed the data into a binary distribution: “0=Pigmented” and “1=Bleached or partially bleached”. The decision was (1) to reduce observer categorisation bias by acknowledging that bleaching status based on pigmentation or percentage area can be subjective; (2) to compare with the 2022 bleaching event (i.e., data in binary form, see below); and (3) to simplify model interpretation. All to obtain consistent bleaching data and reduce methodological bias (Rivera-Sosa et al. 2025).

In addition to the transects performed during 2024, we also analysed previously published data documenting the 2022 bleaching event from Chung et al. (2024). We considered the same sites, Bluff Island (22.3254° N, 114.3543° E), Sharp Island (22.3637° N, 114.2913° E), and Tung Ping Chau (22.5467° N, 114.4311° E) at their respective averaged depths reported in the publication: shallow (2–4 m) and deep (4–6 m). These data also consisted of transects and bleaching categories (for details, see Chung et al. (2024)). The addition of 2022 survey data was intended to increase the robustness of results found in the 2024 surveys and add broader applicability. However, the 2022 surveys were less precise in providing exact depths, so depth-specific and taxonomic analyses focused only on the 2024 bleaching event.

Data exploration and frequentist statistical analyses

Transect data were initially analysed for statistical differences using analysis of variance (package “stats”). The ANOVAs or KRUSKAL.TESTS based on Shapiro test of the residuals were run on the number of colonies

per site to test the effects of transects that were not significant (p -value = 0.826; > 0.05 despite violating normality due to the number of 0 s—see [Bayesian Modelling](#)) and the coral colony size, which was also not significant (AOV p -value = 0.281; > 0.05 residuals p -value = 0.04, but Kruskal–Wallis chi-squared p -value = 0.25). Neither the effects of transects nor colony size were significant in the categorisation of bleaching statuses (AOV p -value = 0.702; > 0.05 residuals p -value = 0.34, and AOV p -value = 0.501; > 0.05 residuals p -value = 0.98, respectively). Then, we tested for statistical similarities in coral community structure using Non-Metric Multidimensional Scaling (NMDS; package “vegan”, Oksanen et al. 2013). The NMDS was computed based on the Bray–Curtis distance dissimilarity matrix on the number of observations per taxa across sites and depths. We then transformed the data, condensing from the four categories (“Pigmented”, “Partially bleached”, “Bleached” and “Dead”) into three (“Pigmented”, “Bleached or partially bleached” and “Dead”) to examine and display the amount of bleaching recovery and mortality using beta regression models (package “betareg”, Cribari-Neto and Zeileis 2010). These were tested as the statistical differences in the proportions (0 to 1 or %) of colonies between the “During” and “After” bleaching surveys and interpreting the log-odds comparisons.

Since we also wanted to analyse bleaching patterns while accounting for taxonomic differences across depths and sites, we used the number of observations (i.e., counts). In order to deal with the overdispersions and lack of normality (with low R^2 and high residuals) (i.e., Poisson distribution), which were compromised because of the large number of zero observations (i.e., no presence or absence) of the less-common corals, we ran a hurdle models (package “pscl” (Zeileis et al. 2008) on the number of observations (i.e., colonies categorised). The hurdle models comprise two stages: the non-zero data (situations in which at least a single coral colony was categorised) and zero-census data (situations where coral colonies were not encountered and categorised at all). These were tested with “Poisson” and “Zero-inflated” distributions to determine the likelihood of having 0 s or no observations across sites and depths. These converged after C24 BFGS (Broyden–Fletcher–Goldfarb–Shanno; i.e., machine learning for smooth optimisation), and concluded that the log odds of having a 0 “no observations” increased with depth (Depth 1 vs 4–5 m, Est = − 0.32, p -value = 0.000736, < 0.05), and that the log odds of having counts “observations” was higher in TPC (Bluff vs TPC Est = 0.4, p -value = 1.52e-05, < 0.05). In other words, more coral colonies and, therefore, more colonies assessed for the bleaching status were found in the shallows than at depth (please note that the difference between 1 vs 2–3 m was not significant, Est = 0.02, p -value = 0.256677). From this point on, we switched the statistical analyses to

Bayesian modelling, carrying only a binomial distribution of “0 = Pigmented” and “1 = Bleached or partially bleached” to (1) explore bleaching patterns with depth and temperature across taxa during the peak of the bleaching, and (2) allow comparison with the 2022 bleaching event.

Bayesian modelling

To explore the relationship between temperatures, depths and bleaching, we fitted several Bayesian Regression Models (Table S1) (packages “brms” and “cmdstanr”; Bürkner 2017). These models fitted different formulas for fixed and random effects, but all considered a Bernoulli (Binary) family and a fixed number of draws 12,000 (9,000 iterations with 3,000 warm-up), tree depth 13 and an adaptive delta 0.99. These fixed sampling parameters were held constant to minimise divergent transitions and ensure that differences in convergence (R-hat and effective sample size) reflected the model outcomes rather than the sampling configuration.

First, we fitted a model for the combined 2022 and 2024 data; the model formula considered “Depth”, “Site” and “Year” as fixed effects and a random intercept and slope for “Taxa” factor with “Depth”. Please note that the survey in situ temperatures were not available for 2022. This model was used to explore the decreasing probability of bleaching with depth and the comparisons across years and sites (see Fig. 2d, e). Secondly, to take advantage of more precise depths and higher taxonomic resolution from the 2024 data, we fitted a second model that converged best with “Temperature” and “Site” as fixed effects and a random intercept and slope for the “Taxa” factor with “Temperature”. The best model accounted for “Temperature” instead of “Depths” or the combination of both (highly correlated), and it was determined with “loo_compare” (package “loo”) after comparing model combinations of “Temperature” and “Depth” from the high-precision DEFI loggers (Table S1). This selection was likely due to collinearity between “Temperature” and “Depth”, which were highly correlated (see data from camera-mounted DEFI, CTDs and long-term in situ loggers). Therefore, we also displayed the results using “Depth” and “Site” as fixed effects and a random intercept and slope for the “Taxa” factor with “Depth” (no “Temperature”).

All ecological analyses were run with R Studio (version 2024.12.0 + 467) with other main packages, including “tidyverse” (Wickham et al. 2019). Temperature maps and figures were run with MATLAB (version R2024a). All codes and data are available for reproducibility (see Data Availability Statement).

Results

Thermal context

Satellite sea surface temperatures (SST)

The SST in this marginal subtropical habitat showed a wide seasonal range, reaching as low as 16.4 °C with maximum temperatures above 30 °C in most years. Temperatures exceeded the BT for 26 out of 39 years of available SST (Fig. 1c), and the duration of heatwaves likely to trigger coral bleaching (SST > BT) has shown an increasing trend in the past decade from < 10 days a year to 37 days in 2024 (Fig. S1).

SST reached a record high in 2022 of 31.0 °C on 26 July 2022 (Fig. 1a), exceeding the BT for a total of 27 days between July and August. This heat led to surface heating accumulation as high as 23.1 °C days on 3 August 2022 (7.1 °C weeks on 31 August 2022), with heating over 8 °C days (as a reference value, see Methods) for 30 days. The maximum SST was slightly lower in 2024, at 30.9 °C on 5 July (Fig. 1b), but SSTs stayed above the BT for longer (37 days), resulting in a heat accumulation above 8 °C days for a total of 46 days and reaching up to 25.2 °C days on 16 July 2024 (Table 1). In 2024, DHW exceeded 8 °C weeks for 20 days, with DHW peaking at 9.24 °C weeks on 15 September 2024.

In situ logger temperatures

Fixed logger temperatures recorded greater temperature variability compared to SST. During both heatwaves, temperatures exceeded 32 °C, with the 2024 maximum (33.21 °C at 2 m at Sharp Island) slightly hotter below the surface than the 2022 maximum (33.15 °C at 3 m at Sharp Island) based on all loggers (Table 1; Fig. 2a). Depth variability was strongly evident in situ, especially during the hottest summer months, with the shallower logger (2–3 m) maximums around 33 °C, hotter than the deepest logger (8 m) maximum of 30.98 °C. These vertical variations were also evident in the DHDs maximums (35.4 °C days at 2 m and 7.9 °C days at 8 m) or the duration above 8 °C days (77 and 0, respectively; Table 1). For example, in the case of Sharp Island in 2024, the maximum difference between the maximum temperatures at 2 m (32.94 °C) and the minimum temperatures at 8 m (23.35 °C) reached 9.59 °C on 12 July 2024. Such daily range across depths can be compared to that during the early dry to wet season transition, when the range was only 1.86 °C (maximum at 2 m = 27.02 °C and minimum at 8 m = 25.16 °C on 30 April 2024). Although spatial variability between sites was less significant than depth variability, loggers also showed some site differences, with Bluff Island ~ 1 °C colder than Sharp Island and Tung Ping Chau (TPC) (Table 1). Finally, consistent with SST,

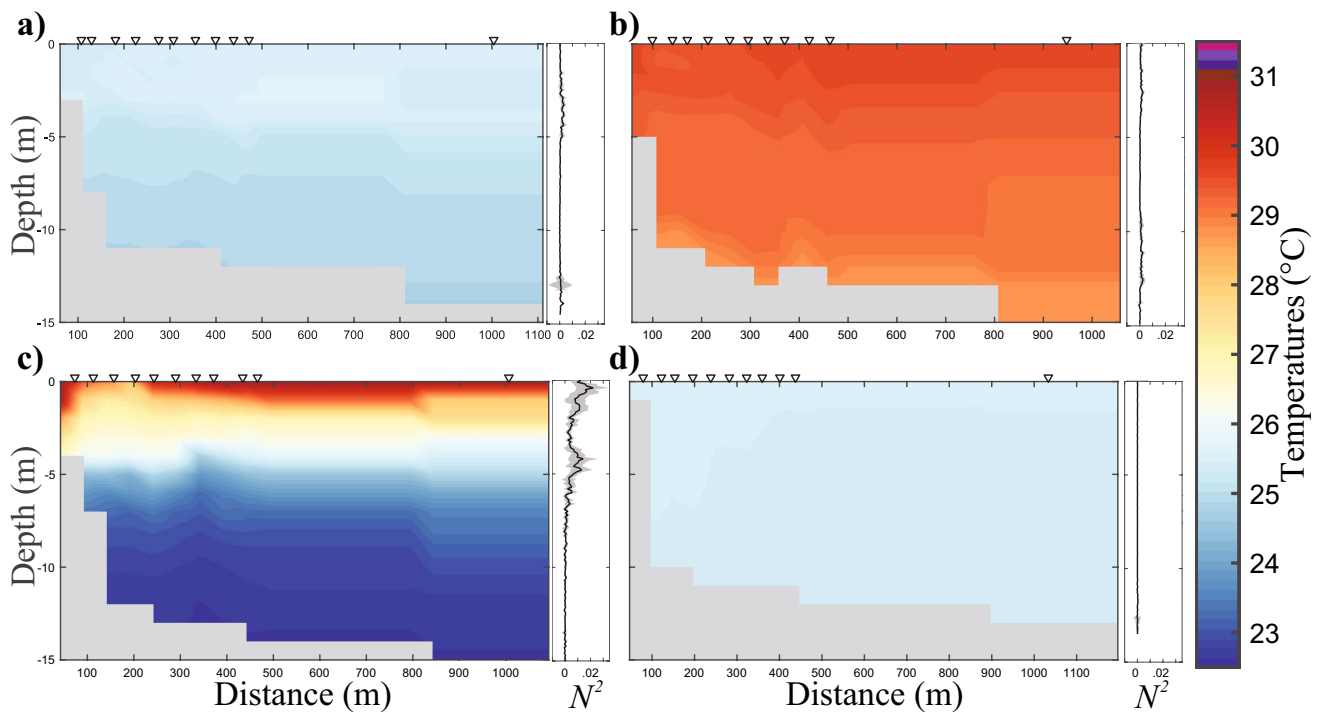


Fig. 3 Seasonality in vertical water temperature along an onshore-offshore transect at Sharp Island. Casts were done on **a** 5 May 2022, **b** 6 October 2022, **c** 6 August 2024, and **d** 12 November 2024 and consisted of 11 CTD vertical profiles (∇) perpendicular to the coast. The right panels for each date show the buoyancy frequency squared

N^2 , highlighting differences in stratification and depth of maximum stratification (max N^2 : 0.0034 at 3.3 m, 0.0025 at 2.6 m, 0.022 at 0.4 m, and 0.00033 at 11.8 m, respectively); the solid line shows the average across casts and the shading the 95% CI. Additional CTD transects for other sites can be found in Fig. S2

heat accumulation in 2024 reached 35.41 DHDs on 17 July 2024 at 2 m at Sharp Island (i.e., just before the bleaching survey), which can be compared to the maximum heat recorded in situ in 2022 of 24.02 DHDs on 23 July 2022 at 3 m at TPC (Table 1). Both heatwaves showed a bimodal pattern, with a rapid reduction in heat accumulation in mid and late August of both 2022 and 2024, followed by a second period of lower heat accumulation lasting until late August for 2022 and the end of September for 2024. These heat accumulation periods were slightly lagged in time (Fig. 2). For 2024, heat accumulated for longer than in 2022 (e.g., max DHD duration $> 8^\circ\text{C}$ days: 20 vs 0 days at 5 m at Bluff Island; 64 vs 39 days at 3 m at Sharp Island; or 58 vs 34 days at 3 m at TPC, respectively, for 2024 and 2022), therefore, accumulating heat for 60% longer or more in 2024 (Table 1; Fig. 2b, c).

Vertical casts

The vertical water column structure from CTD casts supported the pronounced summer stratification evident from the in situ fixed logger temperatures (Fig. 3; Fig. S2). Taking

Sharp Island as an example, on 9 August 2024 (i.e., at the peak of the heatwave and during the time of the bleaching surveys), temperatures varied from 31°C at the surface to 23°C at ~ 15 m. By contrast, on 12 November 2024, temperatures only varied from 25.54 to 25.38°C over the same water column depths. These observations of stratification are quantitatively supported by buoyancy frequency squared (N^2) values (Fig. 3). The maximum N^2 values from CTD casts were one to two orders of magnitude higher in the summer than in winter. In August 2024, these peaked at 0.4 m (0.022 s^{-2}) at Sharp Island (Fig. 3c), 2.8 m (0.019 s^{-2}) at Bluff Island (Fig. S2a), and 1.3 m (0.030 s^{-2}) at TPC (Fig. S2c).

The summer stratification led to observations of bottom temperatures in August 2024, the coldest among the casts reported for Sharp Island, which were not available during the 2022 or 2024 winter peak (i.e., January–February). The CTD casts at Bluff Island and TPC also exhibited similar patterns of summer stratification and winter mixing. On 6 and 9 August 2024, temperatures at Bluff Island ranged from 30.2 to 23.9 and TPC from 31.3 to 22.8°C , respectively, from surface to bottom. Conversely,

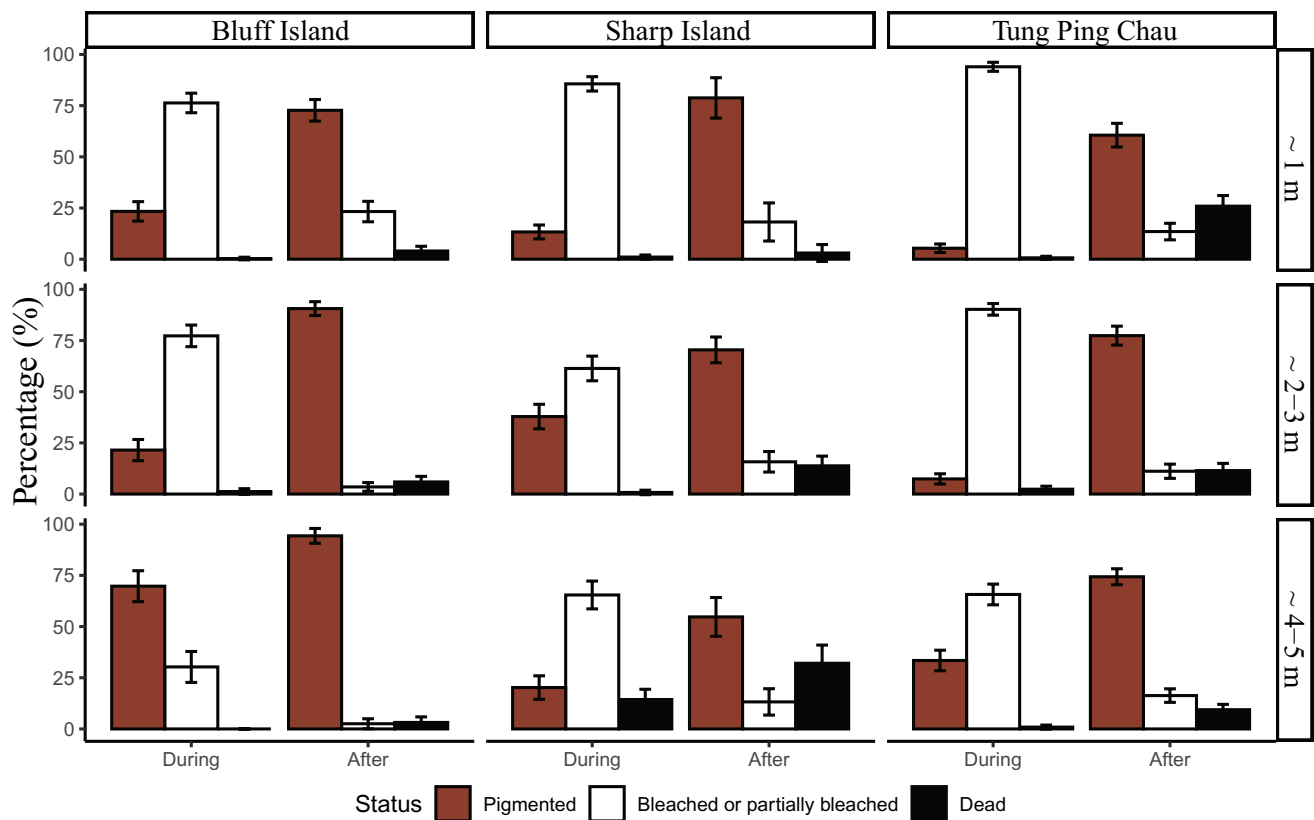


Fig. 4 Proportions of “Pigmented”, “Bleached or partially bleached”, and “Dead” coral colonies at each sampling time, site and depth. Error bars show standard errors using 95% CI. The exact depths are available in Table 2

on 12 November 2024, Bluff Island ranged from 25.48 to 25.38 °C and TPC from 27.15 to 25.5 °C. The same occurred in a complimentary CTD cast at TPC on 11 October 2022, when temperatures ranged from 28.26 to 27.88 °C (Fig. S2). For comparison with summer values, in November 2024, N^2 values indicated well mixed conditions across depths with very small peaks: 0.0003 s^{-2} at 11.8 m for Sharp Island (Fig. 3d), 0.0003 s^{-2} at 0.3 m for Bluff (Fig. S2b), and 0.0017 s^{-2} at 0.7 m for TPC (Fig. S2d). A similar seasonal variation occurred in 2022, although no CTD casts were performed during the peak of the heatwave that year; on 25 May 2022, water temperatures varied from 25.9 to 25.6 °C and on 6 October 2022, from 29.65 to 28.78 °C (Fig. 3).

Spatial and temporal frequentist comparisons of bleaching

Comparing “during” vs. “after” bleaching, post-bleaching analyses showed that most colonies recovered (i.e., indicated by the increase of “Pigmented” colonies), from 25.8% during bleaching to 74.9% after bleaching across depths (beta-reg log-odds = 2.18, p -value = 2.65×10^{-15} , < 0.05). The proportion of “Bleached or partially bleached” colonies

decreased between the two surveys (log-odds = -3.02 , p -value = $< 2 \times 10^{-16}$; < 0.05) (Fig. 4). For example, at 2–3 m, bleaching percentages dropped from 60–90% “during” the bleaching to 4–16% “After” (Bluff Island: 77.3 ± 5.3 (s.e.) to 3.5 ± 2.1 , Sharp Island: 61.3 ± 6 to 15.8 ± 5 , and TPC: 90.1 ± 2.8 to 11.1 ± 3.5). The bleaching percentages at 1 m (Bluff Island: 76.3 ± 4.8 , Sharp Island: 85.6 ± 3.5 and TPC: 93.9 ± 2.2) were higher than at 4–5 m (Bluff Island: 30.3 ± 7.5 , Sharp Island: 65.4 ± 6.8 and TPC: 65.7 ± 5) (beta-reg log-odds = -1.26 for 4–5 m vs. 1–2 m; p -value = 2.92×10^{-5} ; < 0.05). Spatially, bleaching was similar (no significant difference) at TPC and Sharp Island (beta-reg log-odds = -0.31 , p -value = 0.29), and higher than Bluff Island (log odds = -0.86 , p -value = 0.003, < 0.05). Mortality of “Dead” colonies occurred mainly at Sharp Island, was low at Bluff Island ($< 2.7\%$ across depths) and decreased with depth at TPC (25.9% at 1 m, 11.5% at 2–3 m, and 9.4% at 4–5 m). However, at Sharp Island mortality increased significantly with depth (3% at 1 m, 13.8% at 2–3 m, and 32.1% at 4–5 m; log-odds = $+2.52$ for 4–5 m vs. 1 m), where “Dead” colonies were already significant at depth (14.4% at 4–5 m) during the first bleaching survey (Fig. 4).

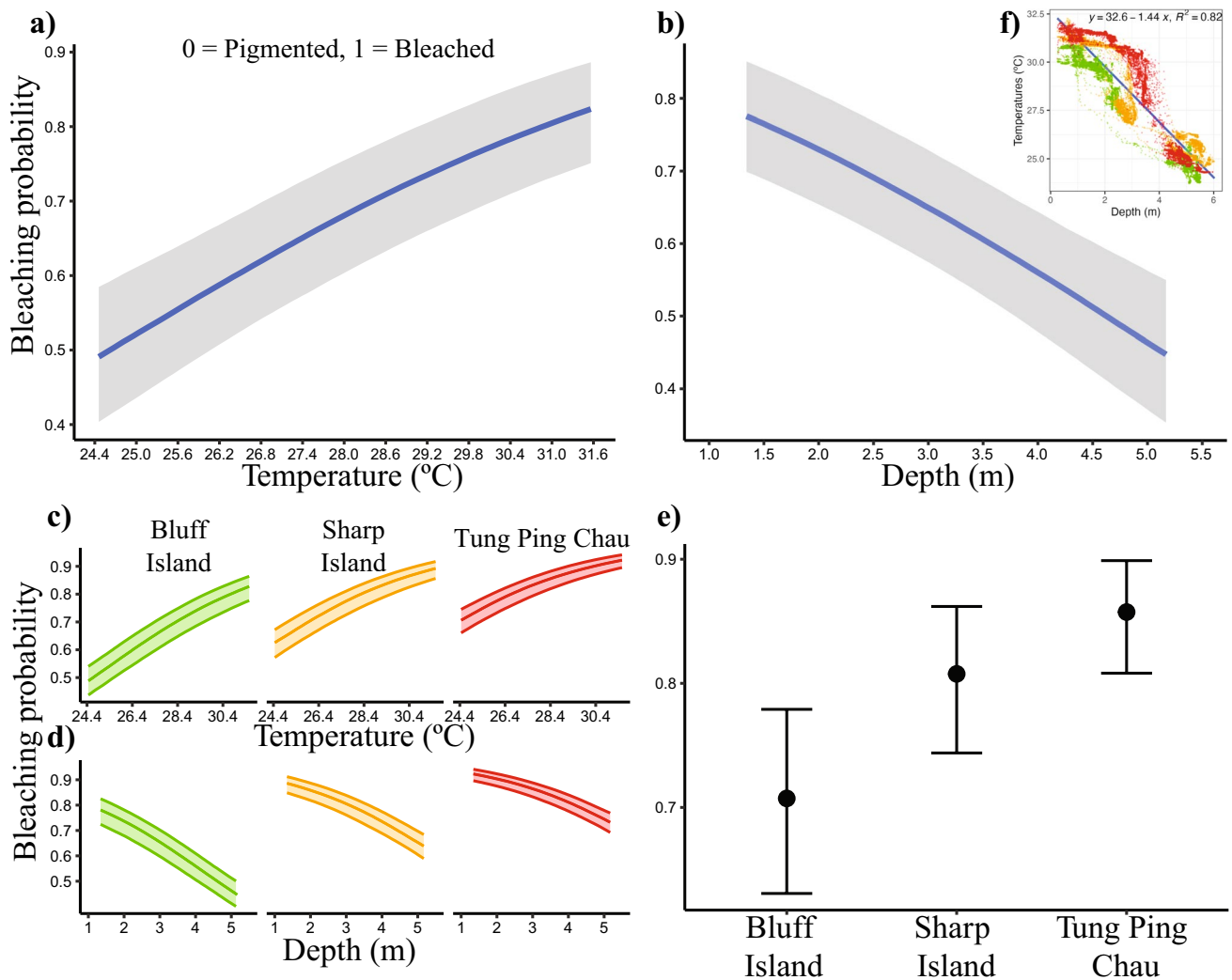


Fig. 5 Bayesian predicted bleaching probability with conditional effects (0=Pigmented, 1=Bleached). Panels display probabilities **a** according to temperatures, **b** according to depths, **c** across sites according to temperatures, **d** across sites according to depth, and **e**

according to sites as a discrete variable. Top-right (**f**) shows depth and temperature collinearities with the correlation equation slope and R^2 . CI 95%

Bayesian bleaching patterns with depth and temperature

For both bleaching events, increasing depths with decreasing temperatures decreased the probability of suffering bleaching (Bayesian Estimate of the slope (Est.) -0.89 with [Confidence Intervals $= -1.41, -0.47$]). Across events, the likelihood of bleaching was higher (Est. 1.58 [$0.96, 2.22$]) in 2024 compared to 2022, and this was consistent across the three sites (Fig. 2d). For both events, the likelihood of bleaching was the lowest at Bluff Island, followed by Sharp Island (Est. 0.69 [$0.27, 1.12$]) and the highest at Tung Ping Chau (Est. 1.16 [$0.73, 1.6$]) (Fig. 2d,e).

For the 2024 event only, the actual in situ temperatures (DEFI) recorded during the bleaching surveys mounted

to the camera frame were a better predictor than depths (Expected Log Pointwise Predictive Density $= -2.0$ and Standard Error Difference $= 1.3$) and converged into the best Bayesian model (Table S1). The likelihood of bleaching increased with increasing temperatures (Est. 0.22 [$0.13, 0.35$]) (Fig. 5a). Following negative collinearity, the correlated decreasing temperatures with increasing depth (Pearson cor -0.906 , p -value < 0.05 ; consistent across individual sites) (Fig. 5f) also caused bleaching prevalence to decrease significantly with depth (Est. -0.39 [$-0.64, -0.20$]) (Fig. 5b). According to the Bayesian model, the likelihood probability of suffering bleaching at around 1 m, where temperatures were close to 31°C , was at 80%, while a few metres deeper at around 5 m (24 – 25°C), the likelihood decreased to below 50% (Fig. 5a, b).

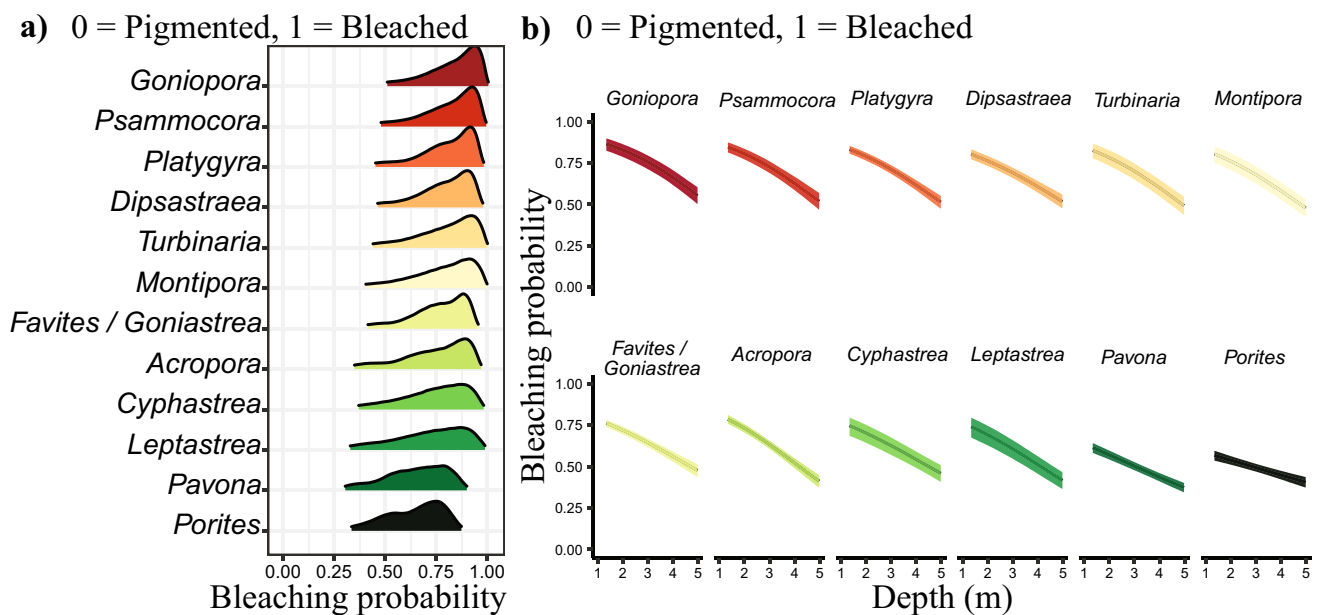


Fig. 6 Bayesian predicted bleaching probability across taxa (0 = Pigmented, 1 = Bleached). **a** Posterior distribution probability of suffering bleaching with taxa ordered (average probability) according to bleaching sensitivity from top to bottom. **b** Probability of bleaching according to depths. Temperatures and depths were highly correlated

Regarding spatial variability, bleaching was slightly more prominent at TPC than Sharp Island and, in turn, than Bluff Island, all along the temperature-depth gradient. These differences were reflected by the Bayesian estimates (at TPC Est. 0.91 [0.48,1.36] and Sharp Island Est. 0.55 [0.11,1] in comparison with Bluff Island) and conditional effects (Fig. 5c, d).

Within genus taxa (visually identified based on taxonomic morphotype), and considering only the 2024 data to maintain maximum taxonomic resolution, we found that all genera decreased their bleaching likelihood with increasing depths (Est. -0.39 [-0.64, -0.20]), which represents an average reduction across sites from 85.3% at ~1 m to 76.3% at ~2–3 m to 53.8% at 4–5 m, correlated with decreasing temperatures (Fig. 5 and 6). Among the most representative taxa for Hong Kong waters (based on literature and genus colonies count; AFCD 2005 and Fig. S4), the most susceptible genus to bleaching was *Goniopora*, with the highest bleaching probability (avg. mean of posterior distribution 0.817). It was followed by *Platygyra* (avg. 0.798) and *Acropora* (avg. 0.731). On the contrary, the least bleaching susceptible taxa were *Porites* (avg. 0.641), followed by *Pavona* (avg. 0.645). At their deepest appearances, around 4–5 m, where temperatures were below 25 °C, all morphotype corals had less bleaching probability than the conspecific (i.e. for taxa) occurrences in the shallows and kept the same susceptibility order (Fig. 6b).

(Pearson cor -0.906, p -value < 0.05). Taxa with fewer than five observations total and classified as “Unidentified” were dropped. All identifications are based on morpho-taxonomy, and voucher examples are available in the Supporting Information

Discussion

Similar to reefs worldwide during the Fourth GCBE (NOAA, 2024; Reimer et al. 2024), the marginal coral assemblages of Hong Kong experienced widespread bleaching during the 2024 heatwave. We found elevated bleaching rates at all sites, which corresponded to maximum observed water temperatures and heat accumulation over several weeks (Table 1). Although Hong Kong’s corals occur in highly challenging thermal regimes or perhaps because of it, we observed that these coral assemblages recovered their pigmentation, indicating potential resilience to thermal stress. Observations of coral mortality were limited despite the severe heatwave and widespread bleaching observations. Summer stratification seemed to play a key role in improving bleaching outcomes, and the level of protection afforded by deeper and cooler water varied between sites. A stratification episode that may have been exacerbated by rainfall (i.e., salinity stratification; Hong Kong Observatory 2024b), potentially increasing surface heating and decreasing temperatures sharply with depth. With significantly colder waters at depth, the likelihood of bleaching was reduced in just a few metres of water. This decrease in bleaching with depth coincides with trends found in many other locations considering larger depth ranges (Baird et al. 2018; Bridge et al. 2014; Crosbie et al. 2019; Goodbody-Gringley & Chequer 2025; MacDonald et al. 2021; Muir et al. 2017; Pérez-Rosales et al. 2021). Our findings support growing

insights that subsurface thermal variability—even across small depth ranges—can influence coral bleaching resilience (Brown et al., 2023; Chung et al. 2024; Green et al. 2019), and such fine-scale temperature variations should be integrated into studies predicting coral vulnerability and persistence under future heatwaves.

Despite considering a narrow depth range, strong thermal gradients were observed. For instance, at Sharp Island, water temperatures cooled from 32 °C at the surface to 23 °C in just 15 m of water during the peak of the heatwave (Fig. 3c). Stratification was most evident in the upper few meters, with maximum N^2 one to two orders of magnitude higher in the summer months, peaking at 2.8 m (0.019 s^{-2}) at Bluff Island (Fig. S2a), 0.4 m (0.022 s^{-2}) at Sharp Island (Fig. 3c), and 1.3 m (0.030 s^{-2}) at TPC (Fig. S2c). These N^2 depth patterns indicate shallow stratification compared with summertime mixed layer depths for other coral reef habitats. Global analysis of ocean exposed reefs suggested mixed layer depths average ~27 m and only shallow to ~16 m, although oscillations of the thermocline, such as due to internal waves, can lead to temporal variability (Wyatt et al. 2020). Such thermocline oscillations might explain the high temperature variability observed with fixed loggers during the summertime (Fig. 2a–c). The strong summer stratification and frequent cooling of deeper corals led to water column temperatures diverging from SST, highlighting the need to consider vertical structure and in situ temperatures at the depth of the corals, especially when aiming to understand the complex relationship between coral bleaching and temperatures with depth. We hypothesise that processes that enhance stratification, such as freshwater inputs and heating at the surface, or lead to vertical mixing, including regular thermocline oscillations and typhoon-associated mixing, may contribute to local variations in bleaching outcomes by relieving or exacerbating coral thermal stress (Heron et al. 2008). The bimodal peaks in DHDs in both 2022 (Fig. 2b) and 2024 (Fig. 2c) show the influence of mixing associated with tropical storms, followed by enhanced stratification, on surface heat accumulation (Zhou et al. 2012). Importantly, while storm-associated mixing might rapidly reduce heat accumulation, our observations suggest rainfall, strong surface heating, and subsequent strong density stratification appear to have contributed to a return to heat stress conditions in both 2022 and 2024 (Fig. 2b, c).

When assessing coral bleaching patterns and outcomes, studies should try to consider variations within genera or species (or at the most precise visual morphotype/taxonomic level possible). Coral species and genera have different bleaching susceptibility (Burn et al. 2023; Conti-Jerpe et al. 2020; Loya et al. 2001; Marshall & Baird 2000; Muir et al. 2021; Yeung et al. 2025) and ignoring the taxa-level effect can lead to inaccurate interpretations. Visually identifying coral species or even genera is challenging, especially

in highly diverse sites or for taxa with close morphological similarities, such as distinguishing species within the genera *Acropora* and *Porites*, or differentiating between the genera *Favites* vs. *Goniastrea* or *Goniopora* vs. *Alveopora* (Chow et al. 2025; Veron et al. 2016, 2025). This approach allowed us to systematically investigate how individual or grouped taxa respond to bleaching and how their bleaching patterns vary with depth (Muir et al. 2021). This level of analysis provides key information for interpreting how dominant community-structuring or minor and highly vulnerable morphotype taxa are affected by bleaching events (Smith 2024).

Hong Kong corals have experienced and are likely to increasingly experience heatwaves triggering coral bleaching (Chung et al. 2024). So far, these bleaching events have not appeared to result in mass mortalities or local extinctions because, although present in small coral communities and numbers, scleractinian corals persist in Hong Kong, especially in eastern areas (i.e., our study sites) (Duprey et al. 2016). We hypothesise that their exposure to highly seasonal temperature ranges and short-period temperature fluctuations during peak heating, sometimes alleviated by rapid vertical mixing, might have acclimatised these corals to thermal stress, as suggested by their ability to recover their symbionts (pigmentation) even after widespread bleaching (Palumbi et al. 2014; Putnam 2021; Safaie et al. 2018). Another potential hypothesis is that due to increased nutrient availability, like particulate organic matter and plankton (Duprey et al. 2016), Hong Kong corals may exhibit resilience enhanced by heterotrophic feeding, which may confer greater bleaching resistance (Conti-Jerpe et al. 2020; Grottoli et al. 2006; Love et al. 2025). Nevertheless, mortality rates were low but present (lower “During” the bleaching survey than “After”; beta-reg log-odds = 1.55, p -value = $8.46\text{e-}05$, < 0.05 , but with no significant effect for “Depth”, p -value = 0.49 of 1 vs 2–3 m and p -value = 0.33 of 1 vs 4–5 m) (Fig. 4). Yet, it was difficult to determine whether mortality was only triggered by bleaching (i.e., partial vs total mortality colonies) or additional stressors in this marginal environment (e.g., sedimentation, hyposalinity, eutrophication and consequent hypoxia, bioerosion) (Cybulski et al. 2020; Dumont et al. 2013; Duprey et al. 2016; Geeraert et al. 2021; Wong et al. 2018; Yeung et al. 2021). For instance, consistent with previous observations (Chung et al. 2024; Qian et al. 2018; Gan et al. 2016), we observed deoxygenation (dissolved oxygen $< 2 \text{ mg/L}$) in bottom waters around Sharp Island (Fig. S5). Such low oxygen levels could potentially explain the higher mortality observed at Sharp Island, especially since coral mortality increased with depth, opposite to the observed thermal stratification pattern. Besides that, mortality was also evident “During” the first survey, likely before thermally induced bleaching mortality (Fig. 4). Consequently, although caution is needed in the interpretation of mortality drivers, our findings suggest that deeper areas of

Hong Kong may be less likely to experience bleaching and consequent mortality during marine heatwaves (Bridge et al. 2014; Frade et al. 2018; Pérez-Rosales et al. 2021). However, this thermal refuge function could be compromised by other stressors typical of marginal reefs, such as deoxygenation, freshwater inputs that trigger eutrophication or sedimentation (i.e., turbidity with limited light penetration) (Duprey et al. 2016; Morgan et al. 2020; Loke et al. 2024; Wong et al. 2018). We suggest incorporating the role of multiple oceanographic drivers, as well as environmental stressors or factors causing coral mortality in Hong Kong, when spatially planning coral conservation efforts.

From a local-scale conservation perspective, among the three studied sites, TPC experienced the highest bleaching (Fig. 4 and 5). This bleaching difference (also observed in 2022) may reflect TPC experiencing consistently hotter temperatures (Table 1 and 2; Fig. 2e and 5), as well as being inhabited by the most diverse coral assemblage (Bray–Curtis distance NMDS stress=0.082; Adonis effect of Site $R^2=0.51$, $F=3.38$, $p\text{-value}=0.008$; Fig. S3), with higher alpha diversity (i.e., richness) (Bluff Island=16, Sharp Island=14 and TPC=19) and a higher abundance of colonies across taxa (Bluff Island=1408, Sharp Island=1197 and TPC=2272) (Fig. 6 and S4) that may be more prone to heat stress. In contrast, Bluff Island experienced less bleaching than Sharp Island and TPC (Fig. 2e, 4e, and S4). Although Bluff and Sharp islands were more similar in terms of community assemblage, Bluff Island experienced slightly colder temperatures and higher oxygen levels than Sharp Island (Table 1 and 2; Fig. 2, 5c,f and S5). Finally, Sharp Island exhibits a continuous decline in corals, which has been attributed to stressors such as bleaching, hypoxia, turbidity, fishing and diving impacts (Chung et al. 2024; So et al. 2023). The present study provides further evidence of significant concerns around hypoxia (Geeraert et al. 2021; Qian et al. 2018; Gan et al. 2016), which could be limiting the thermal stratification refuge potential and making Sharp Island's corals even more vulnerable during heatwaves.

Conclusions

To summarise, our study demonstrates that even modest increases in depth with marked local stratification can substantially reduce bleaching and mortality risk during marine heatwaves in the marginal coral assemblages of Hong Kong. These findings underscore the importance of integrating fine-scale vertical and spatial thermal profiles—based on in situ data—alongside transparent taxonomic assessments to accurately evaluate coral bleaching outcomes. Our results highlight the importance of local oceanographic contexts, including interacting stressors, in modulating bleaching impacts during extreme events, with stratification acting

as a key driver of vertical thermal refuges and a potential mechanism of thermal resilience. We, therefore, advocate for a more holistic, oceanographically informed framework for evaluating bleaching outcomes—one that complements coral biology and strengthens our ability to identify, predict, and safeguard coral communities most likely to persist under increasingly frequent and severe marine heatwaves, such as those leading to the Fourth Global Coral Bleaching Event.

Acknowledgements Fieldwork was performed under permit '(27) in AF MPD 09_3 Pt.25' from the Hong Kong Agriculture, Fisheries and Conservation Department (AFCD). We thank Lucia Hu and boat captains Chun Wai Wong and Vincent Tong for their field assistance and Eszter Matrai (Tursiopsdivers) for our early conversations. The work was funded by the Research Grants Council (RGC) of Hong Kong (RGC Grant #: AoE/P-601/23-N). GP-R was supported by the RGC Postdoctoral Fellowship Scheme (Project No. HKUST PDFS2425-6S04). YDP, JBW, and TBK were supported by the RGC Hong Kong PhD Fellowship Scheme (HKPFS PF22-74876, PF21-66378, and PF19-35625, respectively).

Author contributions Gonzalo Pérez-Rosales: Conceptualisation (lead), Data Curation (lead), Formal Analysis (lead), Investigation (lead), Methodology (lead), Project Administration (supporting), Validation (lead), Visualisation (lead), Writing—Original Draft Preparation (lead), Writing—Review & Editing (lead). Yu-De Pei: Data curation (supporting), Investigation (supporting), Project Administration (supporting), Writing—Review & Editing (supporting). Joshua Bennett-Williams: Data curation (supporting), Investigation (supporting), Writing—Review & Editing (supporting). Timothy B. King: Data curation (supporting), Investigation (supporting), Writing—Review & Editing (supporting). Markus Rummel: Data curation (supporting), Investigation (supporting), Writing—Review & Editing (supporting). Tzu Hao Chung: Data curation (supporting), Writing—Review & Editing (supporting). Alex S.J. Wyatt: Conceptualization (supporting), Data curation (supporting), Funding Acquisition (lead), Investigation (supporting), Methodology (supporting), Project Administration (lead), Resources (lead), Supervision (lead), Validation (supporting), Visualization (supporting), Writing—Review & Editing (supporting).

Funding Open access funding provided by Hong Kong University of Science and Technology, Research Grants Council, University Grants Committee, AoE/P-601/23-N.

Data availability All temperature and coral data with the necessary codes to replicate our analyses are available at: https://github.com/gonzalopr/HongKong_Bleaching_Oceanography. Satellite sea surface temperature (SST) observations can be accessed from the NOAA CRW (coralreefwatch.noaa.gov/satellite/index.php). All data can also be obtained from Dryad (<https://doi.org/10.5061/dryad.ht76hdxrc>).

Declarations

Conflict of interests The authors declare no competing interests.

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